

Integral traces recover arithmetic data without erasing the source fibres

Global denominator return

$$\begin{aligned} x_1, \dots, x_n &\in \mathbf{Q}^\times \\ s_1, \dots, s_{n-1} &\in \mathbf{Z}, \quad \sum_i x_i^{-1} = m/b \end{aligned}$$



Newton identities + exact valuations

$$\begin{aligned} &\text{one reduced denominator } d \\ &\gcd(d, b) = 1, \quad d^n \mid (n-1)!m \end{aligned}$$



$m \mid 4, n \geq 3$: every denominator prime is excluded

$$d = 1: \text{all } x_i \in \mathbf{Z}$$

Necessary boundary: at $n = 2$, $(p/2, p/2)$ has integral first trace for odd p .

Exact trace fibres on one complete raw gate

